Set Theory Midterm Solutions

1. Prove that $(P \Rightarrow Q) \iff \neg(P \wedge \neg Q)$ is true for all P and Q.

Solution

Here is the truth table:

P	Q	$P \Rightarrow Q$	$\neg Q$	$P \wedge \neg Q$	$\neg(P \wedge \neg Q)$
0	0	1	1	0	1
0	1	1	0	0	1
1	0	0	1	1	0
1	1	1	0	0	1

The third and last columns are the same, showing the equivalence.

2. Let G be a graph and B a subclass of its domain. Let $G(B)$ denote the class

$$G(B) = \{y \mid x \in B, (x, y) \in G\}.$$

Show that if $B \subset C$ are subclasses of the domain of G, then $G(B) \subset G(C)$.

Solution

Let $B \subset C$ and let $y \in G(B)$. Since $y \in G(B)$, there is some $x \in B$ such that $(x, y) \in G$. As $B \subset C$, we have $x \in C$. So by the definition of $G(C)$, $y \in G(C)$. Thus $G(B) \subset G(C)$.

3. Show that if G is an equivalence relation on A, then $G = G \circ G$.

Solution

Let $(x, y) \in G$. Because G is reflexive, $(y, y) \in G$, and so $(x, y) = (x, y) \circ (y, y) \in G \circ G$. Thus $G \subset G \circ G$. On the other hand, let $(x, y) \in G \circ G$. Then there is some z in G such that $(x, z) \in G$ and $(z, y) \in G$. So by the transitivity of G, $(x, y) \in G$. So $G \circ G \subset G$.

4. Let $A = \{a, b, c, d, e, f\}$, and let G and H be the following equivalence relations on A:

$$\begin{aligned} G &= I_A \cup \{(a, b), (b, a), (b, c), (c, b), (a, c), (c, a), (d, e), (e, d)\} \\ H &= I_A \cup \{(b, c), (c, b)\} \end{aligned}$$

What are the sets A/H and G/H?

Solution

A/H is the set $\{H_x \mid x \in A\} = \{\{a\},\{b,c\},\{d\},\{e\},\{f\}\}$ and G/H is the following equivalence relation on A/H: $I_{A/H} \cup \{(\{a\},\{b,c\}),(\{b,c\},\{a\}),(\{d\},\{e\}),(\{e\},\{d\})\}$.

5. Let $f : B \to C$ be a function. Prove that f is injective if and only if for every pair of functions $g : A \to B$ and $h : A \to B$, we have that $f \circ g = f \circ h$ implies $g = h$.

Solution

(Only if): Assume that f is injective and let $f \circ g = f \circ h$ for some pair of functions g and h. To show that $g = h$ we show that $g(a) = h(a)$ for any $a \in A$. Indeed, let $a \in A$. Then

$$\begin{aligned} f(g(a)) &= f \circ g(a) \\ &= f \circ h(a) \text{ because } f \circ g = f \circ h \\ &= f(h(a)). \end{aligned}$$

As f is injective, this implies that $g(a) = h(a)$, as needed.

(If): Assume that for every pair of functions $g : A \to B$ and $h : A \to B$, we have that $f \circ g = f \circ h$ implies $g = h$. Let $b, b' \in B$ be such that $f(b) = f(b')$. We show that $b = b'$, this will give us that f is injective.
Let $K_b, K_{b'} : A \to B$ be the constant functions with images b and b' respectively. Then for any a in A,

$$f \circ K_b(a) = f(b) = f(b') = f \circ K_{b'},$$

so $f \circ K_b = f \circ K_{b'}$. By our assumption then $K_b = K_{b'}$. But then $b = K_b(a) = K_{b'}(a) = b'$, (using any $a \in A$.) This is what we needed to show.

6. Let $\{A_i\}_{i \in I}$ be a family of sets, and $B_i \subset A_i$ for each $i \in I$. (*You only have to do part (c) of the following, but doing parts (a) and (b) will help you do it, and get you partial marks if you can't.*)

   (a) Define the product $\prod_{i \in I} A_i$.

   (b) Define the projection $p_j : (\prod_{i \in I} A_i) \to A_j$.

   (c) Show that $\prod_{i \in I} B_i \subset \bigcap_{i \in I} \overline{\overline{p}}_j$. (Where p_i is defined in part (b).)

Note

Sorry, in part (c) I should have had $\overline{\overline{p}}_i(B_i)$ in place of just $\overline{\overline{p}}_j$. Some of you took this to be $\overline{\overline{p}}_i(A_i)$ (at least implicitly). The statement is true for this too, though the converse wouldn't be.

Solution

(a) $\prod A_i = \prod_{i \in I} A_i$ is the set of functions

$$\{a : I \to \cup A_i | a(i) \in A_i (\forall i \in I)\}.$$

We let $a_i = a(i)$.

(b) The projection $p_j : \prod A_i \to A_j$ is the function defined by $p_j(a) = a_j$ for any $a \in \prod A_i$.

(c) Let a be in $\prod A_i$. As $p_i(a) = a_i$, a is in $\overline{\overline{p}}_i(B_i)$ if and only if a_i is in B_i. So a is in $\bigcap_{i \in I} \overline{\overline{p}}_i$ if and only if $a_i \in B_i$ for all $i \in I$. That is, if and only if $a \in \prod_{i \in I} B_i$. (We've proved not just inclusion, but set equality.)